Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 5 times with different random seeds. The learning rate was set to 0.001 with the Adam optimizer.

2 Results

Figure 1 shows the training loss.

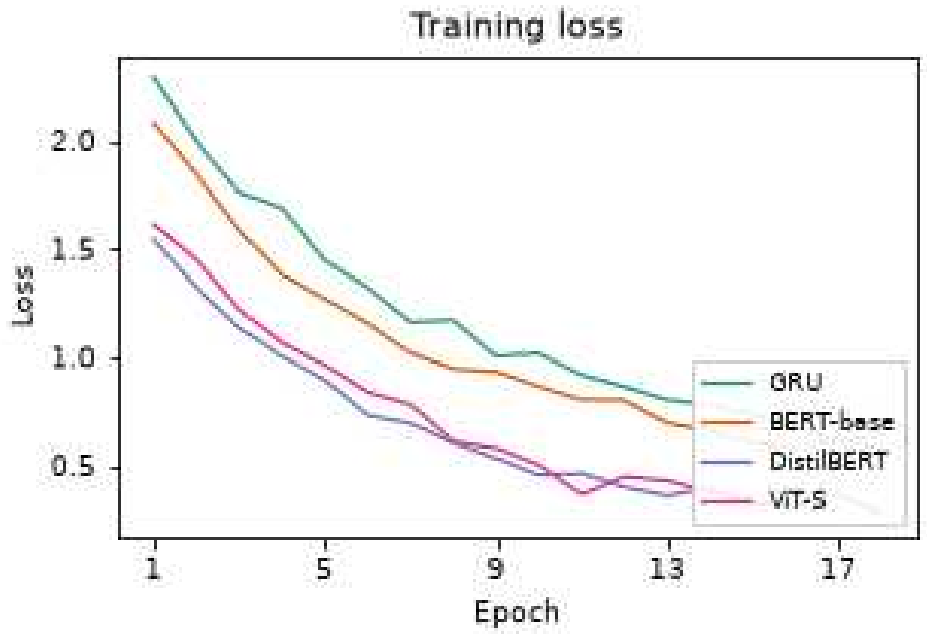


Figure 1: Training loss for the models in this experiment.

At epoch 13, the loss of GRU is over 0.7. As shown in Figure 1, GRU has a loss of 0.338 at epoch 5. At epoch 10, the loss of ViT-S is 1.613. ViT-S has a loss of -0.044 at epoch 4 (Figure 1).

Figure 2 shows the f1 score by dataset.

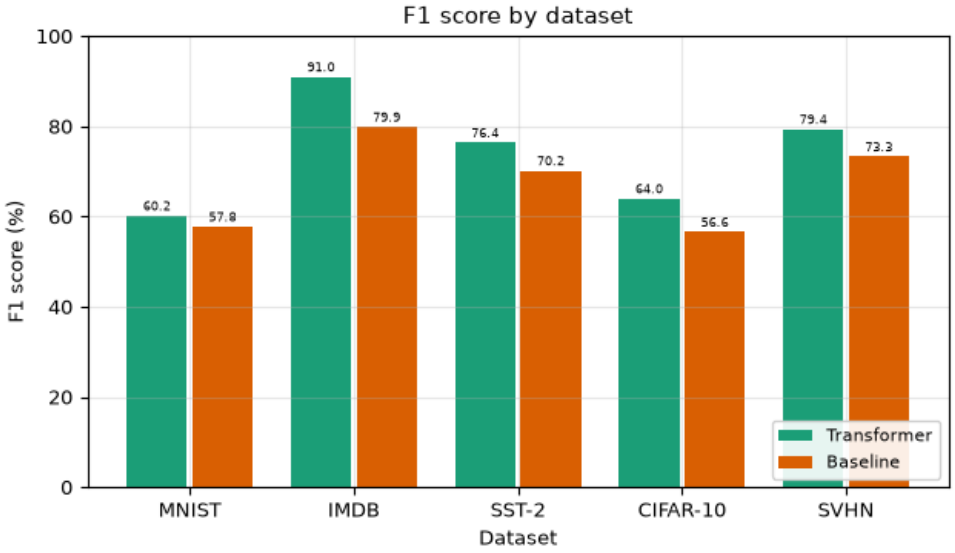


Figure 2: F1 score by dataset for the models in this experiment.

Transformer outperforms Baseline by 12.4 percentage points on MNIST. Transformer achieves 77.4% F1 score on SVHN. As shown in Figure 2, Transformer achieves 64.0% F1 score on CIFAR-10. Transformer reaches 76.4% F1 score on SST-2.

Figure 3 shows the accuracy during training.

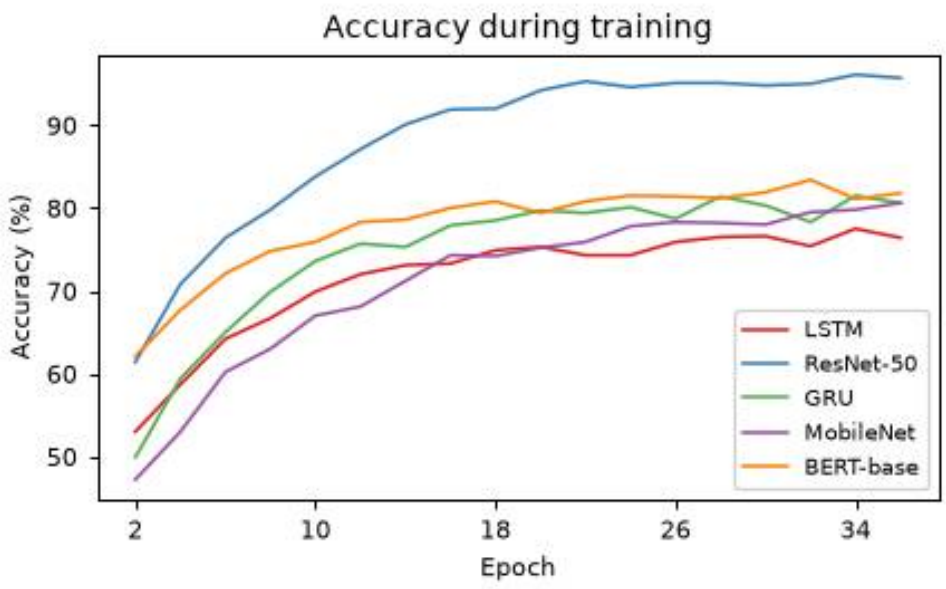


Figure 3: Accuracy during training for the models in this experiment.

At epoch 32, MobileNet achieves 79.5% accuracy. As shown in Figure 3, BERT-base outperforms GRU by 11.0 percentage points at epoch 4.

Figure 4 shows the accuracy by dataset.

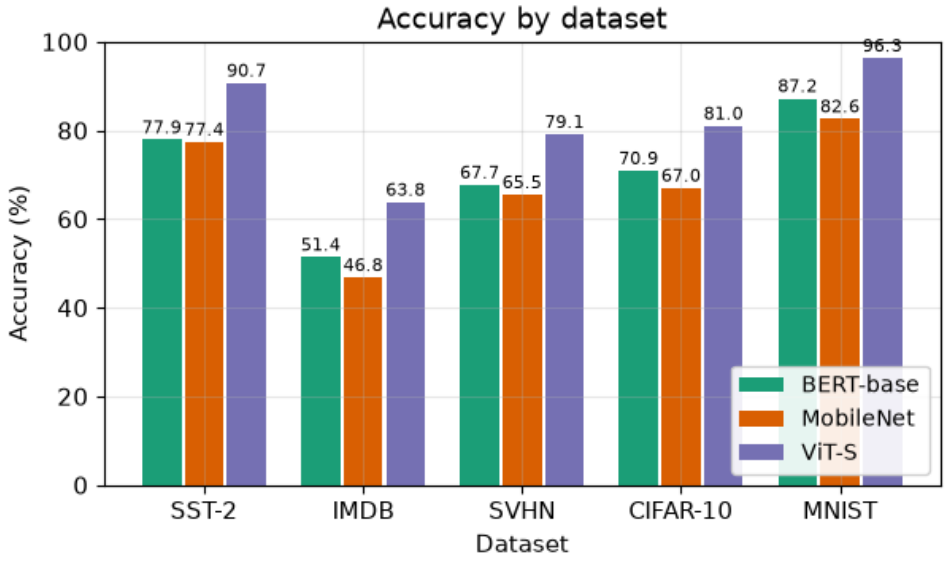


Figure 4: Accuracy by dataset for the models in this experiment.

As shown in Figure 4, MobileNet reaches 77.4% accuracy on SST-2. As shown in Figure 4, BERT-base outperforms MobileNet by 1.9 percentage points on CIFAR-10. ViT-S outperforms BERT-base by 32.8 percentage points on SST-2. BERT-base outperforms MobileNet by 52.2 percentage points on SVHN.

3 Discussion

The differences between the models are discussed in the next section.